

DRM: Directional Relational Manifolds

Relational Metric, Transport, Curvature, and Conditional Toroidal Closure

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Revised version, July 2026

Abstract

We introduce Directional Relational Manifolds (DRM), a geometric framework in which dimensionality is not assumed to be globally fixed, but emerges from the rank and interaction structure of locally active relational directions.

A DRM does not require all local directional sets to be embedded in a single pre-existing metric space. Instead, each state possesses its own local relational vector space, relational metric, anchor into the state space, and transport law along admissible trajectories. Continuity is therefore defined internally through relational transport rather than through an external Hausdorff metric.

The relational metric is allowed to be positive semidefinite. Its kernel represents collapsed or inactive directions, while its rank defines the effective local dimensionality. A directional connection is constructed from relational transport, and curvature is defined through infinitesimal non-commutativity or finite-loop holonomy.

We further clarify the relation between recurrence and toroidal topology. Boundedness and recurrence alone do not imply that an attractor is a torus. Toroidal closure follows only under additional assumptions, such as the existence of independent commuting recurrent phase flows with a full-rank period lattice.

Under regularity, constant-rank, non-degeneracy, and compatibility conditions, DRM reduces to Riemannian geometry. When the relational metric is the Fisher Information Matrix, it also reduces to Fisher–Rao information geometry. DRM is therefore proposed as a variable-rank relational extension of classical differential and information geometry.

1 Introduction

Classical differential geometry generally begins with a smooth manifold of fixed dimension. Each point is locally modeled on the same Euclidean space $\mathbb{R}^n$, and the tangent bundle has constant rank.

This structure is extremely successful for systems whose effective degrees of freedom remain stable. However, many adaptive systems exhibit a different behavior:

- active degrees of freedom may emerge or disappear;
- directional modes may become degenerate;
- interactions may be strongly non-orthogonal;
- the effective state space may change with the state itself;
- local geometry may be produced by relations rather than imposed before them.

Directional Relational Manifolds are introduced to formalize this possibility.

The central principle of DRM is:

Geometry does not necessarily precede the system. Geometry may emerge from the active relations of the system.

Accordingly, dimensionality is not defined by counting an arbitrary set of directions. It is defined by the effective rank of the relational structure.

2 Conceptual Motivation

Euclidean and Riemannian geometry usually assume:

1. a fixed local dimension;
2. a pre-existing tangent structure;
3. a metric defined on tangent vectors;
4. a common differentiable structure across the state space.

DRM relaxes these assumptions.

The local relational spaces need not all have the same dimension. A direction can become active, inactive, redundant, or degenerate. The metric itself may lose rank, and this loss of rank is interpreted as a genuine change in the effective dimensionality of the system.

The geometry is therefore not merely a passive arena in which dynamics occurs. The geometry is part of the dynamics.

3 Relational State Space

A DRM requires enough structure to define trajectories and local directions, but it does not require a single Euclidean ambient space containing all directional sets.

Definition 3.1 (Relational State Space). *A relational state space is a Hausdorff topological space M equipped with a specified class C of admissible continuous curves*

$$\gamma : I \rightarrow M,$$

where $I \subseteq \mathbb{R}$ is an interval.

Whenever differentiability is required, M is assumed to possess a compatible smooth, stratified-smooth, or differential-space structure along the curves in C .

Remark 3.1. *The framework does not require M to be a globally fixed-dimensional smooth manifold. It is sufficient that derivatives can be defined along admissible trajectories or inside regular strata.*

4 Local Relational Spaces

Definition 4.1 (Local Relational Space). *For every $p \in M$, let $\mathcal{E}_p$ be a real vector space called the local relational space at p .*

The spaces $\mathcal{E}_p$ are not initially assumed to form a vector bundle of constant rank.

Definition 4.2 (Active Directional Set). *For each $p \in M$, let*

$$\mathcal{D}(p) \subseteq \mathcal{E}_p$$

be a set of active relational directions.

The associated active linear space is

$$\mathcal{E}_p^{\text{act}} = \text{span}_{\mathbb{R}} \mathcal{D}(p).$$

Thus, $\mathcal{D}(p)$ is not treated as a vector space by itself. It is a generating set contained in the local relational vector space $\mathcal{E}_p$.

Definition 4.3 (Anchor Map). *An anchor at p is a linear map*

$$\rho_p : \mathcal{E}_p^{\text{act}} \rightarrow T_p M$$

whenever a tangent space $T_p M$ is defined.

The anchor maps internal relational directions to observable state-space velocities.

Remark 4.1. *The anchor need not be injective. Different internal directions may produce the same observable motion. It need not be surjective either, since not every formal state-space direction must be dynamically accessible.*

5 Relational Metric

Definition 5.1 (Directional Relational Metric). *A directional relational metric is a family*

$$g_p : \mathcal{E}_p^{\text{act}} \times \mathcal{E}_p^{\text{act}} \rightarrow \mathbb{R}$$

such that, for every $p \in M$:

1. *g_p is bilinear;*
2. *g_p is symmetric:*

$$g_p(v, w) = g_p(w, v);$$

3. *g_p is positive semidefinite:*

$$g_p(v, v) \geq 0.$$

The value $g_p(v, w)$ measures relational coupling between active directions. It need not imply that the directions are orthogonal or spatial in the Euclidean sense.

Definition 5.2 (Relational Kernel). *The relational kernel at p is*

$$\mathcal{N}_p = \text{Ker } g_p = \{v \in \mathcal{E}_p^{\text{act}} \mid g_p(v, w) = 0 \text{ for every } w \in \mathcal{E}_p^{\text{act}}\}.$$

Directions in $\mathcal{N}_p$ are metrically invisible at p . They may represent:

- inactive directions;
- collapsed degrees of freedom;
- redundant modes;
- gauge-like directions;
- unresolved internal variation.

Definition 5.3 (Effective Relational Space). *The effective relational space at p is the quotient*

$$\bar{\mathcal{E}}_p = \mathcal{E}_p^{\text{act}} / \mathcal{N}_p.$$

The metric g_p induces a positive-definite inner product $\bar{g}_p$ on $\bar{\mathcal{E}}_p$ whenever the quotient is finite dimensional or otherwise admits the required completion.

Definition 5.4 (Emergent Dimensionality). *The effective DRM dimensionality at p is*

$$\dim_{\mathcal{D}}(p) = \text{rank}(g_p).$$

Equivalently, in the finite-dimensional case,

$$\dim_{\mathcal{D}}(p) = \dim \bar{\mathcal{E}}_p.$$

Remark 5.1. *This definition replaces the expression*

$$\dim_{\mathcal{D}}(p) = |\mathcal{D}(p)|.$$

Cardinality does not generally measure dimension. A set may contain infinitely many elements while generating a one-dimensional subspace.

6 Relational Transport

Ordinary Hausdorff continuity cannot be used to compare $\mathcal{D}(p)$ and $\mathcal{D}(q)$ unless both sets are embedded in a common metric space.

DRM does not assume such an external ambient metric. Comparison between local relational spaces is instead performed through transport.

Definition 6.1 (Relational Transport). *Let $\gamma : [a, b] \rightarrow M$ be an admissible curve.*

A relational transport along γ is a family of linear maps

$$P_{s \rightarrow t}^{\gamma} : \mathcal{E}_{\gamma(s)}^{\text{act}} \rightarrow \mathcal{E}_{\gamma(t)}^{\text{act}}, \quad s, t \in [a, b],$$

satisfying:

$$P_{t \rightarrow t}^{\gamma} = \text{Id},$$

and, whenever composition is defined,

$$P_{u \rightarrow t}^{\gamma} \circ P_{s \rightarrow u}^{\gamma} = P_{s \rightarrow t}^{\gamma}.$$

The composition law may be weakened in approximate or discrete implementations, but the exact law is assumed in the formal continuous model.

Axiom 6.1 (Internal Relational Continuity). *Let γ be an admissible curve and let v, w be relational fields along γ .*

The DRM is relationally continuous along γ when

$$t \mapsto g_{\gamma(t)}(v(t), w(t))$$

is continuous whenever $v(t)$ and $w(t)$ are transported continuously by P^{γ} .

A stronger metric-compatibility condition is:

$$g_{\gamma(t)}(P_{s \rightarrow t}^{\gamma} v, P_{s \rightarrow t}^{\gamma} w) = g_{\gamma(s)}(v, w). \quad (1)$$

This equality is not required in every DRM. When it holds, the transport is metric-preserving. When it fails, the defect measures relational deformation.

Definition 6.2 (Metric Deformation). *The metric deformation along γ between s and t is*

$$\Delta_{s,t}^{\gamma}(v, w) = g_{\gamma(t)}(P_{s \rightarrow t}^{\gamma} v, P_{s \rightarrow t}^{\gamma} w) - g_{\gamma(s)}(v, w).$$

Thus, the geometry can evolve dynamically without requiring an external metric space in which all directional sets already live.

7 Directional Connection

A directional connection should not merely be named. It should be constructed from the relational transport.

Definition 7.1 (Relational Field Along a Curve). *A relational field along an admissible curve γ is a map*

$$V(t) \in \mathcal{E}_{\gamma(t)}^{\text{act}}.$$

Definition 7.2 (Directional Covariant Derivative). *Let $V(t)$ be a relational field along γ . The directional covariant derivative of V along γ is*

$$\frac{D^{\mathcal{D}}V}{dt}(t) = \lim_{h \rightarrow 0} \frac{P_{t+h \rightarrow t}^{\gamma} V(t+h) - V(t)}{h},$$

whenever this limit exists.

If the curve velocity is generated by a relational field $U(t)$ through

$$\dot{\gamma}(t) = \rho_{\gamma(t)}(U(t)),$$

we may write

$$\nabla_U^{\mathcal{D}} V = \frac{D^{\mathcal{D}}V}{dt}.$$

Axiom 7.1 (Linearity). *For relational fields U, V, W and scalars $a, b \in \mathbb{R}$,*

$$\nabla_U^{\mathcal{D}}(aV + bW) = a\nabla_U^{\mathcal{D}}V + b\nabla_U^{\mathcal{D}}W.$$

Axiom 7.2 (Leibniz Rule). *For a scalar function f ,*

$$\nabla_U^{\mathcal{D}}(fV) = \rho(U)[f]V + f\nabla_U^{\mathcal{D}}V.$$

Definition 7.3 (Metric-Compatible DRM Connection). *The connection is metric-compatible when*

$$\rho(U)[g(V, W)] = g(\nabla_U^{\mathcal{D}}V, W) + g(V, \nabla_U^{\mathcal{D}}W).$$

Metric compatibility is optional in the most general DRM, since some systems may dynamically generate or dissipate relational strength.

8 Directional Curvature

Curvature measures the failure of relational transport to be path-independent.

8.1 Infinitesimal Curvature

Whenever a bracket $[U, V]$ is defined for relational fields, define

$$R^{\mathcal{D}}(U, V)W = \nabla_U^{\mathcal{D}}\nabla_V^{\mathcal{D}}W - \nabla_V^{\mathcal{D}}\nabla_U^{\mathcal{D}}W - \nabla_{[U, V]}^{\mathcal{D}}W.$$

This quantity measures the infinitesimal non-commutativity of directional transport.

8.2 Finite-Loop Curvature

Let ℓ be a closed admissible loop based at p .

Definition 8.1 (Relational Holonomy). *The relational holonomy of ℓ is*

$$\text{Hol}_\ell^{\mathcal{D}} = P_{0 \rightarrow 1}^\ell : \mathcal{E}_p^{\text{act}} \rightarrow \mathcal{E}_p^{\text{act}}.$$

Definition 8.2 (Loop Curvature Operator). *The finite-loop curvature operator is*

$$\mathcal{R}_\ell^{\mathcal{D}} = \text{Hol}_\ell^{\mathcal{D}} - \text{Id}.$$

If

$$\mathcal{R}_\ell^{\mathcal{D}} \neq 0,$$

then a relational direction transported around the loop does not return unchanged.

This definition is especially natural for DRM because it does not require all local spaces to be embedded in a common global vector space.

9 Admissible Dynamics

Definition 9.1 (Admissible Curve). *A differentiable curve $\gamma : I \rightarrow M$ is admissible when there exists a relational field $V(t) \in \mathcal{E}_{\gamma(t)}^{\text{act}}$ such that*

$$\dot{\gamma}(t) = \rho_{\gamma(t)}(V(t)).$$

Definition 9.2 (Relational Energy). *The relational energy of the pair (γ, V) is*

$$\mathcal{E}[\gamma, V] = \frac{1}{2} \int_I g_{\gamma(t)}(V(t), V(t)) dt,$$

subject to

$$\dot{\gamma}(t) = \rho_{\gamma(t)}(V(t)).$$

Definition 9.3 (Relational Length). *The relational length is*

$$L[\gamma, V] = \int_I \sqrt{g_{\gamma(t)}(V(t), V(t))} dt.$$

Definition 9.4 (Relational Geodesic). *A relational geodesic is an admissible pair (γ, V) that is stationary for the relational energy under admissible variations with fixed endpoints.*

Path uniqueness is not assumed.

Several relational fields may generate the same observable velocity through the anchor. Moreover, degenerate directions may produce families of energetically equivalent paths.

10 Directional Dynamics

Let $\mathcal{P}$ denote a configuration space of DRM states, metrics, active directions, anchors, or transport laws.

A DRM evolution may be represented abstractly by

$$\dot{x} = F(x), \quad x \in \mathcal{P}.$$

Depending on the application, x may encode:

$$x = (M, \mathcal{E}, \mathcal{D}, g, \rho, P).$$

The local dimension evolves according to

$$\dim_{\mathcal{D}}(p, t) = \text{rank } g_{p,t}.$$

A dimensional transition occurs when the rank changes:

$$\text{rank } g_{p,t^-} \neq \text{rank } g_{p,t^+}.$$

Such transitions may represent:

- emergence of a new active degree of freedom;
- collapse of a previously active direction;
- synchronization between directions;
- redundancy;
- phase transition;
- cognitive or physical loss of accessible states.

11 Boundedness, Recurrence, and Stability

The following properties may be imposed on a DRM dynamical realization.

Definition 11.1 (Bounded Orbit). *An orbit $\{\Phi^t(x)\}_{t \geq 0}$ is bounded when it is contained in a compact subset of the configuration space.*

Definition 11.2 (Recurrence). *A point x is recurrent when, for every neighborhood U of x , there exists a sequence $t_k \rightarrow \infty$ such that*

$$\Phi^{t_k}(x) \in U.$$

Definition 11.3 (Structural Stability). *A flow is structurally stable on an invariant set K when sufficiently small perturbations generate flows that are topologically equivalent on K .*

Remark 11.1. *Boundedness, recurrence, and structural stability do not, by themselves, imply toroidal topology.*

A bounded recurrent invariant set may have a topology different from a torus. Therefore, toroidal closure requires additional phase-structure assumptions.

12 Conditional Toroidal Closure

Toroidal topology is naturally associated with multiple independent recurrent phases.

A sphere describes a globally constrained closed geometry, but it does not automatically represent several independent cyclic phases. By contrast,

$$\mathbb{T}^n = \underbrace{S^1 \times \dots \times S^1}_{n \text{ factors}}$$

directly represents n independent recurrent angular variables.

Hypothesis 12.1 (Independent Recurrent Phase Hypothesis). *Let $\mathcal{A} \subseteq \mathcal{P}$ be a compact, connected, invariant DRM attractor.*

Assume that $\mathcal{A}$ admits n complete vector fields

$$X_1, \dots, X_n$$

such that:

1. *the fields commute:*

$$[X_i, X_j] = 0 \quad \text{for all } i, j;$$

2. *they are pointwise linearly independent on $\mathcal{A}$;*

3. *their joint flow defines a transitive action*

$$\Psi : \mathbb{R}^n \times \mathcal{A} \rightarrow \mathcal{A};$$

4. *the stabilizer of one, and therefore every, point is a full-rank lattice*

$$\Lambda \subset \mathbb{R}^n.$$

Theorem 12.1 (Conditional Toroidal Closure). *Under the Independent Recurrent Phase Hypothesis,*

$$\mathcal{A} \cong \mathbb{R}^n / \Lambda.$$

Since every full-rank lattice quotient of $\mathbb{R}^n$ is diffeomorphic to an n -torus,

$$\mathcal{A} \cong \mathbb{T}^n.$$

Proof. The commuting complete vector fields generate an action of the additive Lie group $\mathbb{R}^n$ on $\mathcal{A}$.

Transitivity implies that $\mathcal{A}$ is diffeomorphic to a homogeneous quotient

$$\mathbb{R}^n / \text{Stab}(p).$$

By assumption,

$$\text{Stab}(p) = \Lambda,$$

where Λ is a discrete full-rank lattice.

Therefore,

$$\mathcal{A} \cong \mathbb{R}^n / \Lambda.$$

Every quotient of $\mathbb{R}^n$ by a full-rank lattice is diffeomorphic to $\mathbb{T}^n$. □

Remark 12.1. *The theorem does not state that every stable DRM converges to a torus.*

It states that a compact DRM attractor possessing n independent commuting recurrent phases with a full-rank period lattice has toroidal topology.

Remark 12.2 (Toroidal Preference). *The torus is preferred over the sphere when the global relational dynamics contains independent recurrent cycles.*

For example, two independent phases naturally generate

$$S^1 \times S^1 = \mathbb{T}^2,$$

rather than

$$S^2.$$

This is not because the sphere is topologically incomplete. The sphere is a complete compact manifold. The distinction is that the torus encodes independent periodic coordinates, whereas the sphere does not possess the same product structure of independent cycles.

13 Reduction to Riemannian Geometry

DRM should recover classical geometry when its variable and relational features become regular and fixed.

Theorem 13.1 (Riemannian Reduction). *Let $(M, \mathcal{E}, \mathcal{D}, g, \rho, P)$ be a DRM satisfying:*

1. *M is a smooth n -dimensional manifold;*
2. *the active spaces $\mathcal{E}_p^{\text{act}}$ form a smooth vector bundle $\mathcal{E} \rightarrow M$ of constant rank n ;*
3. *the anchor*

$$\rho_p : \mathcal{E}_p^{\text{act}} \rightarrow T_p M$$

is a vector-space isomorphism for every p ;

4. *g_p is smooth and positive definite;*
5. *the induced connection is metric-compatible and torsion-free.*

Then the DRM structure induces a Riemannian metric on M , and its connection coincides with the Levi–Civita connection.

Proof. Because ρ_p is an isomorphism, define a metric h on TM by

$$h_p(X, Y) = g_p(\rho_p^{-1}X, \rho_p^{-1}Y).$$

The smoothness and positive definiteness of g imply that h is a smooth Riemannian metric.

The induced DRM connection transfers through ρ to a metric-compatible, torsion-free connection on TM .

By uniqueness of the Levi–Civita connection, the transferred DRM connection is the Levi–Civita connection of h . $\square$

Remark 13.1. *Orthonormality of the original active set $\mathcal{D}(p)$ is not required.*

A non-orthogonal generating set may define the same Riemannian metric, provided the induced bilinear form is positive definite and the anchor is an isomorphism.

14 Reduction to Information Geometry

Let

$$\mathcal{S} = \{p(x; \theta) \mid \theta \in \Theta \subseteq \mathbb{R}^n\}$$

be a regular parametric statistical model.

The Fisher Information Matrix is

$$g_{ij}^F(\theta) = \mathbb{E}_{p(x; \theta)} [\partial_i \log p(x; \theta) \partial_j \log p(x; \theta)].$$

Corollary 14.1 (Fisher–Rao Reduction). *Suppose a DRM satisfies:*

1. $\dim_{\mathcal{D}}(p) = n$ at every regular point;
2. the anchor identifies

$$\bar{\mathcal{E}}_p \cong T_{\theta} \mathcal{S};$$

3. the effective relational metric satisfies

$$\bar{g}_p(e_i, e_j) = g_{ij}^F(\theta);$$

4. the DRM connection is the metric-compatible torsion-free connection induced by $\bar{g}$.

Then the effective DRM geometry is the Fisher–Rao statistical manifold.

Proof. The constant-rank condition produces a regular n -dimensional effective relational space.

The anchor identifies this space with the tangent space of the statistical model. The relational metric becomes the Fisher Information Matrix.

By the Riemannian reduction theorem, the induced metric-compatible, torsion-free connection is the Levi–Civita connection associated with the Fisher–Rao metric. $\square$

Remark 14.1. *DRM extends information geometry when one or more of the following occurs:*

- the Fisher-like metric changes rank;
- the effective number of parameters varies with state;
- relational directions become degenerate;
- internal directions are not globally identifiable with parameter coordinates;
- transport modifies the relational metric;
- different strata possess different effective dimensions.

15 Physical, Cognitive, and Computational Interpretation

In a physical system, $\mathcal{E}_p$ may represent locally available modes of motion or interaction.

In a cognitive system, $\mathcal{E}_p$ may represent active reasoning directions, memory pathways, hypotheses, or internal transformations.

In an artificial intelligence system, $\mathcal{E}_p$ may represent the currently accessible relational state transitions of the model.

The kernel

$$\mathcal{N}_p = \text{Ker } g_p$$

may represent temporarily inaccessible, redundant, synchronized, or collapsed directions.

A change in

$$\text{rank } g_p$$

then represents a measurable change in effective dimensionality.

This allows DRM to describe systems in which the state space is not merely traversed, but continuously reconstructed by the activity of the system.

16 Discussion

The corrected DRM formalism makes four central changes.

16.1 Directions Are Locally Typed

The active set $\mathcal{D}(p)$ is no longer treated as a structureless collection from which curvature can be directly calculated.

Instead,

$$\mathcal{D}(p) \subseteq \mathcal{E}_p,$$

where $\mathcal{E}_p$ is a local relational vector space.

This makes linear combinations, kernels, ranks, metrics, and transport well-defined.

16.2 Continuity Is Internal

DRM does not require an external metric space containing every $\mathcal{D}(p)$.

Therefore, ordinary Hausdorff continuity is not assumed.

Continuity is expressed through relational transport and the behavior of observable relational quantities such as

$$g_{\gamma(t)}(V(t), W(t)).$$

16.3 Connection Is Constructed

The directional connection is defined from the infinitesimal behavior of the transport maps:

$$\nabla^{\mathcal{D}} \sim \lim_{h \rightarrow 0} \frac{P_{t+h \rightarrow t} - \text{Id}}{h}.$$

Curvature then follows from transport around loops or from non-commuting covariant derivatives.

16.4 Toroidal Topology Is Conditional

Recurrence alone does not prove toroidal topology.

The torus follows when recurrence is organized into independent commuting phases with a discrete full-rank period lattice.

Thus, toroidal topology remains a central DRM hypothesis, but it is no longer claimed as a consequence of boundedness and recurrence alone.

17 Conclusion

Directional Relational Manifolds provide a framework in which geometry is generated by active relations rather than imposed as a fixed background.

Each state p possesses a local relational space $\mathcal{E}_p$, an active generating set $\mathcal{D}(p)$, a positive-semidefinite relational metric g_p , an anchor ρ_p , and a transport law along admissible trajectories.

The effective local dimensionality is

$$\dim_{\mathcal{D}}(p) = \text{rank } g_p.$$

Degenerate directions belong to the metric kernel and are removed from the effective geometry through the quotient

$$\bar{\mathcal{E}}_p = \mathcal{E}_p^{\text{act}} / \text{Ker } g_p.$$

Relational continuity is defined through internal transport rather than an external Hausdorff distance. The directional connection is constructed from transport, while curvature is measured by infinitesimal commutators or finite-loop holonomy.

Toroidal topology is not inferred from recurrence alone. It arises conditionally when the invariant DRM dynamics possesses independent commuting recurrent phases whose periods form a full-rank lattice.

Under constant-rank, smoothness, non-degeneracy, anchor-isomorphism, and compatibility conditions, DRM reduces to Riemannian geometry. When the metric is the Fisher Information Matrix, DRM reduces to Fisher–Rao information geometry.

DRM therefore proposes a relational and variable-rank extension of classical geometry for adaptive physical, cognitive, informational, and computational systems.

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